

Supplementary material: Continuous multisensory proprioception model

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1 Optimal control model details

1.1 General equations for LQG control

The state of the environment $\mathbf{x}_t$ evolves according to a discrete-time linear dynamical system with additive Gaussian noise:

$$\mathbf{x}_{t+1} \sim \mathcal{N}(\mathbf{A}\mathbf{x}_t + \mathbf{B}\mathbf{u}_t, \mathbf{\Sigma}_x), \quad (1)$$

with noise covariance $\mathbf{\Sigma}_x$.

At each time step, the agent receives a noisy observation $\mathbf{y}_t$ of the state.

$$\mathbf{y}_t \sim \mathcal{N}(\mathbf{H}\mathbf{x}_t, \mathbf{\Sigma}_y), \quad (2)$$

The observation is a linear transformation of the state plus additive Gaussian noise with covariance $\mathbf{\Sigma}_y$.

The cost function is quadratic in the state and in the action:

$$\sum_{t=1}^T \mathbf{x}_t^\top \mathbf{Q} \mathbf{x}_t + \mathbf{u}_t^\top \mathbf{R} \mathbf{u}_t. \quad (3)$$

This control problem is known as linear-quadratic-Gaussian (LQG) control. Its optimal solution combines optimal state estimation via the Kalman filter (KF) (Kalman, 1960a) with optimal feedback control via the linear-quadratic regulator (LQR) (Kalman, 1960b).

The KF iteratively computes the posterior distribution over the state given observations, $p(\mathbf{x}_t \mid \mathbf{y}_1, \dots, \mathbf{y}_t) = \mathcal{N}(\hat{\mathbf{x}}_t, \mathbf{\Sigma}_t)$. The mean state estimate and its covariance are updated as

$$\hat{\mathbf{x}}_{t+1} = \mathbf{A}\hat{\mathbf{x}}_t + \mathbf{B}\mathbf{u}_t + \mathbf{K}_{t+1}(\mathbf{y}_{t+1} - \mathbf{H}(\mathbf{A}\hat{\mathbf{x}}_t + \mathbf{B}\mathbf{u}_t)) \quad (4)$$

$$\mathbf{\Sigma}_{t+1} = (\mathbf{I} - \mathbf{K}_{t+1}\mathbf{H})\mathbf{P}_{t+1}, \quad (5)$$

where $\mathbf{K}_t$ is

$$\mathbf{K}_{t+1} = \mathbf{P}_{t+1} \mathbf{H}^\top (\mathbf{H} \mathbf{P}_{t+1} \mathbf{H}^\top + \Sigma_y)^{-1}, \quad (6)$$

$$\mathbf{P}_{t+1} = \mathbf{A} \Sigma_t \mathbf{A} + \Sigma_x. \quad (7)$$

$$(8)$$

The linear-quadratic regulator yields a feedback policy expressed in terms of the current state estimate:

$$\mathbf{u}_t = \mathbf{L}_t \hat{\mathbf{x}}_t, \quad (9)$$

This feedback gain minimizes expected cumulative cost via dynamic programming; see e.g. Kochenderfer et al. (2022) for derivations of $\mathbf{L}_t$ and $\mathbf{K}_t$.

1.2 Target and hand dynamics

We model the subject’s left hand as a point mass (position, velocity, and force states), while the target (right hand or visual target) follows a random walk with variance σ_{rw}^2 . The model state is

$$\mathbf{x}_t = \begin{bmatrix} x_t^{\text{right}} \\ x_t^{\text{left}} \\ v_t^{\text{left}} \\ f_t^{\text{left}} \end{bmatrix}. \quad (10)$$

The left-hand dynamics are given by a point-mass model commonly used to describe reaching movements (see e.g. Kasuga et al., 2022, for details). Concretely, the continuous-time dynamics for the hand’s velocity and muscle force are

$$m \dot{v}^{\text{left}} = -g v^{\text{left}} + f^{\text{left}}, \quad (11)$$

$$\tau \dot{f}^{\text{left}} = -f^{\text{left}} + u, \quad (12)$$

where g is a damping constant, m is the mass, τ is the muscle time constant, and u is the motor command. Together with $\dot{x}^{\text{left}} = v^{\text{left}}$ these form a linear continuous-time system with matrices $\mathbf{A}_c$ and $\mathbf{B}_c$. Discretizing with time step dt (matrix exponential or zero-order hold) yields the discrete-time matrices $\mathbf{A}_{\text{pm}}$ and $\mathbf{B}_{\text{pm}}$. The continuous-time process noise is transformed to the discrete-time covariance Σ_{pm} via Van Loan’s method (Van Loan, 2003).

The target (right hand or visual target) is modeled as a Gaussian random walk with variance σ_{rw}^2 :

$$x_{t+1}^{\text{right}} = x_t^{\text{right}} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_{\text{rw}}^2). \quad (13)$$

Stacking up the dynamics for the target and the left hand, the full discrete-time dynamics for the whole state are therefore

$$\mathbf{A} = \text{diag}(\mathbf{I}_1, \mathbf{A}_{\text{pm}}), \quad (14)$$

$$\mathbf{B} = \text{diag}(0, \mathbf{B}_{\text{pm}}), \quad (15)$$

$$\Sigma_x = \text{diag}(\sigma_{\text{rw}}^2, \Sigma_{\text{pm}}). \quad (16)$$

The cost function penalizes squared tracking error between response and target positions and squared control signals. The state and action cost matrices are

$$\mathbf{Q} = \text{diag} \left(500 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, 0, 0 \right), \quad (17)$$

$$\mathbf{R} = [c], \quad (18)$$

where c scales the action cost. The factor 500 in $\mathbf{Q}$ was chosen so that c remains in a numerically convenient range during fitting; since only the relative weighting of state and action costs matters, this does not change the model qualitatively.

1.3 Delayed multisensory observation model

We model sensory delays by augmenting the state vector (cf. Izawa & Shadmehr, 2008) and model multisensory observations as independent noisy channels for each sensory modality (cf. Crevecoeur et al., 2016; Kasuga et al., 2022).

Each sensory observation is the signed difference between right and left hand positions. Each modality $k \in \{\text{prop}, \text{vis}\}$ has its own delay δ_k and noise variance σ_k^2 :

$$\mathbf{y}_t^{(k)} = \mathbf{H}\mathbf{x}_{t-\delta_k} + \mathbf{w}_t^{(k)}, \quad \mathbf{w}_t^{(k)} \sim \mathcal{N}(\mathbf{0}, \Sigma_{\mathbf{y}}^{(k)}), \quad (19)$$

$$\mathbf{H} = \begin{bmatrix} 1 & -1 & 0 & \dots \end{bmatrix}, \quad \Sigma_{\mathbf{y}}^{(k)} = [\sigma_k^2]. \quad (20)$$

The complete sensory signal received by the agent is

$$\mathbf{y}_t = [\mathbf{y}_t^{\text{prop}}, \mathbf{y}_t^{\text{vis}}]^\top. \quad (21)$$

We set $\delta_{\text{vis}} = 2$ and $\delta_{\text{prop}} = 1$, which at our frame rate $dt = 0.075$ s corresponds to sensory delays of 150 ms (vision) and 75 ms (proprioception). The noise variances are free parameters and are estimated from the data.

To handle sensory delays, we augment the latent state by stacking previous states up to the maximum delay $\delta_{\text{max}} = \max_k \delta_k$ and adjust the cost and dynamics matrices following Izawa and Shadmehr (2008) such that the previous δ_{max} states are stored in the state vector, but do not affect the dynamics or cost function:

$$\tilde{\mathbf{x}}_t = [\mathbf{x}_t, \mathbf{x}_{t-1}, \dots, \mathbf{x}_{t-\delta_{\text{max}}}]^\top, \quad (22)$$

$$\tilde{\mathbf{x}}_{t+1} = \tilde{\mathbf{A}}\tilde{\mathbf{x}}_t + \tilde{\mathbf{B}}\mathbf{u}_t + \tilde{\mathbf{v}}_t, \quad (23)$$

$$\tilde{\mathbf{Q}} = \text{blkdiag}(\mathbf{Q}, \mathbf{0}, \dots, \mathbf{0}), \quad (24)$$

$$\tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{I} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{I} \end{bmatrix}. \quad (25)$$

Each modality k then uses an augmented observation matrix $\tilde{\mathbf{H}}^{(k)}$ that selects the appropriately delayed component from $\tilde{\mathbf{x}}_t$ (see Crevecœur et al., 2016):

$$\mathbf{y}_t^{(k)} = \tilde{\mathbf{H}}^{(k)} \tilde{\mathbf{x}}_t + \mathbf{w}_t^{(k)}, \quad (26)$$

$$\tilde{\mathbf{H}}^{(k)} = [\mathbf{0} \quad \dots \quad \mathbf{0} \quad \mathbf{H} \quad \mathbf{0} \quad \dots \quad \mathbf{0}], \quad (27)$$

where the block $\mathbf{H}$ appears in the column block corresponding to delay δ_k , and all other blocks are zero.

1.4 Model fitting

We fit the model to behavioral data using inverse optimal control (IOC) for LQG systems (Straub & Rothkopf, 2022). IOC defines the likelihood of a trajectory of observed states conditional on model parameters, $p(\mathbf{x}_{1:T}^{(i)} | \boldsymbol{\theta})$, where $\mathbf{x}_{1:T}^{(i)}$ denotes the T -step sequence of observed left- and right-hand positions in trial i . Individual trials are assumed independent and identically generated by the model with parameters $\boldsymbol{\theta}$, so the dataset likelihood is $p(\mathcal{D} | \boldsymbol{\theta}) = \prod_i p(\mathbf{x}_{1:T}^{(i)} | \boldsymbol{\theta})$.

To model the unisensory and multisensory conditions we assume the same action cost c and action variability σ_{action} across conditions. We also assume a single proprioceptive uncertainty parameter σ_{prop} and two visual uncertainty parameters, $\sigma_{\text{vis},1}$ and $\sigma_{\text{vis},2}$, one for each visual stimulus level.

Overall, the model has five free parameters

$$\boldsymbol{\theta} = \{c, \sigma_{\text{action}}, \sigma_{\text{prop}}, \sigma_{\text{vis},1}, \sigma_{\text{vis},2}\},$$

where $\sigma_{\text{vis},1}$ and $\sigma_{\text{vis},2}$ are the visual noise parameters for the two visual stimulus levels.

We use weakly informative priors for the parameters:

$$c \sim \text{HalfNormal}(1)$$

$$\sigma_{\text{action}} \sim \text{HalfNormal}(1)$$

$$\sigma_{\text{prop}} \sim \text{HalfNormal}(20)$$

$$\sigma_{\text{vis},1}, \sigma_{\text{vis},2} \sim \text{HalfNormal}(20)$$

To test for optimal integration of visual and proprioceptive information we consider several model variants that differ only in how the multisensory condition is modeled; the unisensory conditions are modeled identically across variants.

1. In the *proprioceptive-only* model, only the proprioceptive sensory modality is used in the multisensory condition.
2. In the *visual-only* model, only the visual sensory modality is used in the multisensory condition.
3. In the *reliability-weighted integration* model, both visual and proprioceptive information is used in the multisensory condition and they are combined optimally as prescribed by a Kalman filter.

4. As a heuristic implementation of an *equal-integration* model, we assume that averaging happens at the level of parameters, such that both sensory modalities use the same parameter $\sigma_i = \sqrt{(\sigma_{\text{vis},i}^2 + \sigma_{\text{prop}}^2)}/2$ in the multisensory condition, while unisensory conditions still use their own modality-specific parameters $\sigma_{\text{vis},i}$ and σ_{prop} , respectively.

1.5 Simulation and evaluation of the calibration phase

1.5.1 Model with visual offset

To model a visual offset in the calibration phase of the experiment, we augmented the latent state with a scalar variable b_t that represents the position of the visual cursor. This variable affects the visual observation but not the proprioceptive observation. If $\mathbf{x}_t^{\text{base}}$ denotes the original latent state for either model, the augmented state is

$$\mathbf{x}_t = \begin{bmatrix} \mathbf{x}_t^{\text{base}} \\ b_t \end{bmatrix}. \quad (28)$$

The proprioceptive observation matrix is

$$\mathbf{H}_{\text{prop}} = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \end{bmatrix}, \quad (29)$$

$$(30)$$

and the visual observation matrix is

$$\mathbf{H}_{\text{vis}} = \begin{bmatrix} 0 & -1 & 0 & 0 & 1 \end{bmatrix}, \quad (31)$$

such that the visual element of the state affects only the visual observation.

We assume that the agent is unaware of the bias, and the agent’s subjective internal model stays as in the original model without bias. The agent therefore does not expect a visual offset and does not maintain a separate belief about the visual cursor state.

1.5.2 Simulation of the calibration phase

To simulate behavior in the calibration phase of the experiment, we took the 100 samples of the parameters from the posterior for each model and each participant. For each level of visual offset in the calibration phase, ranging from 0 to 5 cm in steps of 0.5 cm, we simulated one trial of tracking behavior per sampled parameter set. We averaged the tracking error (difference between left hand and visual target) across the 100 simulated trials to obtain the predicted tracking error for each level of visual offset.

1.5.3 Evaluation of the calibration phase

To evaluate predictive performance, we evaluated the log-likelihood of the observed behavior in the calibration phase under the model with parameters sampled from the posterior (100 samples per model and participant).

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